Finite recurrences with exact Fock structure and a zero-tail selection theorem for the helium ground state

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We propose the ansatz $e^{-\sigma/2}\sum_k \ln^k\sigma \sum_i g_i F_{k,i}$ for the helium ground state in Pekeris' perimetric coordinates, where $\sigma = 2\varepsilon(r_1+r_2)$, the $F_{k,i}$ are triple Laguerre series and the g_i are transcendental "atoms" taken from the exact Fock coefficients. Through Fock order three the Laguerre coefficients obey an explicit finite recurrence: at $E=-\varepsilon^2$, $Z=2$, the five unweighted order-three blocks have 2 577 274 stencil entries and shell half-bandwidth at most 16. Separately, we prove that the zero-tail backward (Miller) solution of the weak recurrence, with the constant Laguerre coefficient fixed to one, converges in form norm to the ground state. Finite-section experiments show undamped propagation of tail data and convergence consistent with algebraic decay; no asymptotic rate or Pincherle-type classification is proved. We distinguish proofs, machine checks, numerical observations and open questions.

I. INTRODUCTION

Pekeris turned the helium Schrödinger equation into a 33-term recurrence for Laguerre coefficients in perimetric coordinates [2, 3, 9]. Each Pekeris function is analytic in the interparticle distances, so it cannot carry Fock's $R^2\ln R$ term at the triple-collision point $R=(r_1^2+r_2^2)^{1/2}=0$ [1, 5]. Frankowski and Pekeris added logarithmic basis functions variationally [4]. Morgan proved convergence of Fock-type expansions near $R=0$ [6]; his abstract leaves open whether the physical eigenfunction is represented by one.

We ask whether the ground state can be a "special function" in a stronger sense: one global expansion with (i) the exact Fock structure at $R=0$ built into the basis, (ii) an exponential factor carrying the decay, (iii) coefficients that obey an explicit finite recurrence, and (iv) the physical solution selected by a criterion stated on that recurrence. We construct (i) and (iii) through Fock order three and prove a selection theorem for (iv) on the physical weak form of the recurrence. Two gaps remain open: a convergent global reconstruction of the restricted sectors, and the identification of that weak form with the formal row-cleared recurrence of (iii). Labels: PROVED (argument in the record [10]), VERIFIED (exact or high-precision machine check), NUMERICAL, OPEN, CONJ.

II. ANSATZ AND LOGARITHMIC SECTORS

With $u=\varepsilon(r_2+r_{12}-r_1)$, $v=\varepsilon(r_1+r_{12}-r_2)$, $w=2\varepsilon(r_1+r_2-r_{12})$ and $\sigma=u+v+w$, set

$$\Psi = e^{-\sigma/2}\sum_{k\ge 0}\ln^k\sigma\sum_i g_i\,F_{k,i}(u,v,w), \tag{1}$$

where $F_{k,i}=\sum A^{(k,i)}_{lmn}L_l(u)L_m(v)L_n(w)$ and $g_0=1$. Write $F_k=\sum_i g_iF_{k,i}$. Each sector is restricted to log-free homogeneous expansions at $R=0$ with $F_k=O(R^{2k})$; without this restriction the split is not unique (the gauge $F_0\to F_0-G\ln\sigma$, $F_1\to F_1+G$). The exponential is a gauge, not the boundary condition, since an unrestricted Laguerre series can undo it. Decay is imposed below as membership in the Coulomb form domain.

Let $P=-(2v+w)(2u+w)(u+v)\,e^{\sigma/2}(H-E)e^{-\sigma/2}$, a second-order operator with polynomial coefficients M_{ij},b_i,c. The product rule for $\ln^k\sigma$ gives, for the plain atom $g_0=1$, the sector equations

$$\sigma^2PF_j+(j+1)C_1F_{j+1}+(j+1)(j+2)C_2F_{j+2}=0, \tag{2}$$

with $C_1=2\sigma\sum_{ij}M_{ij}\partial_j+\sigma\sum_i b_i-\sum_{ij}M_{ij}$ and

$$\begin{aligned}C_2=\sum_{ij}M_{ij}&=4\varepsilon^2(u+v)(2u+w)(2v+w)\\&=128\,\varepsilon^5r_1r_2r_{12},\end{aligned} \tag{3}$$

proportional to the physical volume density.

Proposition 1 (VERIFIED)**.** *Every operator in (2) acts on Laguerre coefficients by a finite stencil. At $E=-\varepsilon^2$ the offsets number 33 for P (Pekeris' recurrence), 177 for σ^2P, 117 for C_1 and 57 for C_2; at general E, P has 57 offsets and σ^2P has 225.*

Two independent derivations agree on all 441 coefficients. The Pekeris part agrees with the Koutschan–Zeilberger transcription of Pekeris' Eq. (22) [9], and direct application of H to random arrays in every sector reproduces the stencils exactly. The coupling is upper-triangular in the log level, but the tower has no top sector, so (2) is not solved from the top down. We use (2) formally; an identity for the infinite tower needs a fixed gauge and enough convergence to differentiate and extract coefficients. At the simple ground eigenvalue a finite tower has no positive log sectors, provided every sector times $e^{-\sigma/2}$ lies in the physical operator domain (PROVED; the triple-point value is nonzero).

III. ATOMS AND THE ORDER-THREE RECURRENCE

At Fock order two the only non-polynomial content in distance coordinates is $Z\chi$, where $R^2\psi_{20}=$ polynomial +

TABLE I. Order-three recurrence for $E=-\varepsilon^2$, $Z=2$. "Sectors" counts atom states, i.e. unknown Laguerre arrays per log level. These are sufficient presentations, not minimal ranks. Row-clearing factors have degree ≤ 12.

Block	Sectors	Offsets	Half-bandwidth
Classical	21	215 406	12
Panels 1, 2 (each)	23	562 570	16
Panels 3, 4 (each)	23	618 364	16
Total	113	2 577 274	16

$ZR^2\chi$ and χ is a Clausen-type function. Let $P_+ = (2R^2 - r_{12}^2)^{1/2} = |\mathbf{x}_1 + \mathbf{x}_2|$, $\alpha = 2\arctan(r_2/r_1)$ and $\beta = \arcsin[(R^2 - r_{12}^2)/R^2]$.

Proposition 2 (order two; PROVED)**.** *The smallest module over rational functions of (r_1, r_2, r_{12}) that contains 1 and χ and is closed under differentiation has rank five, with basis $\{1,\ L = \ln\frac{s+r_{12}}{R},\ \alpha,\ V = \frac{P_+}{r_{12}}(\beta + \frac{\pi}{2}),\ U\}$, $s = r_1 + r_2$, where U is a weight-two combination of Cl_2 values and single logarithms. Independence follows from monodromy. One analytic expression covers both orderings $r_1 \gtrless r_2$. The cleared log-sector recurrence is finite, with shell half-bandwidth 10 at $E = -\varepsilon^2$.*

The $1/r_{12}$ pole of V prevents form admissibility even after exponential damping; $r_{12}V$ removes it. Exponentially damped polynomial products of $1, \tilde{L}, \alpha - \pi/2, r_{12}V, U$ are form-admissible, with $\tilde{L} = L - \frac{\pi-2}{\pi}\ln(s/R)$ for the $\ln\sigma$ convention (PROVED).

At order three, $R^3\psi_{30}$ decomposes into a polynomial, a classical block (Clausen functions and logarithms) and four elliptic "panels", which are moments N_0, N_2 of an algebraic integrand over a quartic. For a cyclic vector of the combined module $\{1, \chi, R^3\psi_{30}\}$, finite-field reconstruction of the connection along a test line fails through total (numerator plus denominator) degree 350; separate panels reconstruct with degrees 47–49 (VERIFIED). The high degree is an artefact of the cyclic basis, so we keep the blocks separate.

Proposition 3 (order three)**.** *(a)* PROVED*: the classical block has an exact* 18*-state connection (its pullback to distance coordinates is real,* VERIFIED*). Each panel has a radical-free* 55*-state connection over $\mathbb{Q}(r_1, r_2, r_{12})$, which reduces to* 23 *reachable states over $\mathbb{Q}(\pi)(r_1, r_2, r_{12})$. The panel atoms are global on the open physical triangle: with h a panel ratio such as r_1/r_2, the two moment kernels satisfy $g_j(h/s)\,h/s^2 = -g_j(s)$, so $\int_1^h g_j = 0$ and N_0, N_2 equal their fixed-endpoint integrals from* 1*. (b)* VERIFIED*: the cleared log-sector Laguerre recurrences of the five unweighted blocks have the sizes in Table I.*

Applying $e^{\sigma/2}(H - E)e^{-\sigma/2}$ directly, with the actual atoms, to random Laguerre arrays in every sector at log levels $0, 1$ reproduces the stencils to a relative 10^{-54} (classical block) and 10^{-30} (panels). An independent symbolic derivation reproduces selected panel-3 blocks exactly (1 062 monomials). A compiled generator regenerates all stencils on demand [10].

Form-domain admissibility. Exponentially damped polynomial products of the classical order-three atoms are form-admissible without weights (PROVED). Some panel primitives are not even in L^2 near collinear configurations. Let $\delta = [4r_1^2r_2^2 - (r_1^2 + r_2^2 - r_{12}^2)^2]/R^4$; this is $16\,\mathrm{Area}^2/R^4$ for the electron–nucleus triangle and vanishes exactly on collinear configurations.

Proposition 4 (PROVED)**.** *Exponentially damped polynomial products of all panel atoms with δ^4 lie in the Coulomb form domain. No power δ^a with $0 \le a < 1$ suffices for every panel primitive. Polynomials times $e^{-\sigma/2}\delta^4$ are already form-dense in the singlet S sector.*

The weight changes the angular jets. Table I counts the unweighted operators; weighted stencils need the $d\ln\delta^4$ correction to the connection. The least sufficient power, and whether $R^3\psi_{30}$ lies exactly in a finite span of weighted atoms, are OPEN.

IV. SELECTION BY ZERO-TAIL BACKWARD RECURSION

At fixed ε let V_N be $e^{-\sigma/2}$ times exchange-symmetric Laguerre polynomials of total shell $l + m + n \le N$. The multiplier in P is, up to a negative constant, the volume density, so the rows of the plain recurrence are the Galerkin rows of $H - E$ in V_N up to normalization and exchange multiplicities. Let $K_N(E)$ be that finite section.

Theorem 5 (zero-tail selection; PROVED)**.** *Assume that E_0 is the isolated simple ground eigenvalue, with positive eigenfunction ψ_0. Then: (i) $\bigcup_N V_N$ is a core for the Coulomb form on the singlet S sector; (ii) the coefficient A_{000} extends to a bounded functional on the form domain, and $A_{000}(\psi_0) > 0$; (iii) for each N the system "$A_{000} = 1$, $A_{lmn} = 0$ above shell N, every row through shell N except the constant row, at $E = E_0$" has a unique solution ψ_N, and $\psi_N \to \psi_0/A_{000}(\psi_0)$ in form norm with a Céa bound $\|\psi_N - \psi\| \le C \inf_{v\in V_N,\, A_{000}(v)=1} \|v - \psi\|$; (iv) where $K_N(E)$ is invertible the dropped row is $g_N(E) = 1/[K_N(E)^{-1}]_{000,000}$; its zero near E_0 is eventually the ground Ritz value, and $g_N(E_0) \to 0$.*

Sketch. (i) A functional annihilating every V_N annihilates $e^{-\sigma/2}e^{\mathbf{t}\cdot\mathbf{X}}$ on the tube $|\mathrm{Re}\,\mathbf{t}| < \varepsilon$, by rotational averaging and holomorphy; Fourier inversion then shows it vanishes. (ii) Hardy's inequality in $\mathbb{R}^6$ bounds A_{000}, and its representing density is positive. (iii) Write $v = t\psi_0 + v_\perp$. On $\ker A_{000}$ the bound $|t| \le \|A_{000}\|\,\|v_\perp\|/A_{000}(\psi_0)$ and the spectral gap make $q - E_0$ coercive, so Céa's lemma applies. (iv) The Schur identity gives the formula; coercivity gives $g_N(E_0) \to 0$, and density, min–max and the nonzero head overlap identify the Ritz zero. □

The theorem extends to nested, form-dense, exchange-symmetric enrichments of V_N by $\ln\sigma$, $\ln^2\sigma$, the corrected

TABLE II. $E - E_0$ (hartree) with basis size in parentheses, polynomial degree N. Columns: plain Laguerre; constant replaced by the exact Fock seed $1 - Z\kappa(r_1^2 + r_2^2 - r_{12}^2)\ln\sigma$, $\kappa = \frac{\pi-2}{3\pi}$, plus $\ln\sigma$ times polynomials; plus $\ln^2\sigma$ times polynomials; plus parity-adapted polynomial products of the admissible order-two atoms (min5); min5+ψ_{30}. These Ritz spaces are not the 113-state construction of Table I.

N	plain	$+\ln$	$+\ln^2$	min5	$+\psi_{30}$
6	4.8e-6	6.2e-8	4.6e-8	3.9e-12	3.9e-12
	(50)	(77)	(86)	(141)	(144)
8	9.9e-7	1.6e-9	6.5e-10	2.8e-16	2.8e-16
	(95)	(158)	(195)	(341)	(354)
10	2.7e-7	6.2e-11	1.2e-11	2.3e-20	1.9e-20
	(161)	(279)	(361)	(660)	(694)
11	1.5e-7	1.4e-11	1.6e-12	1.2e-21	7.3e-22
	(203)	(357)	(469)	(873)	(923)

order-two atoms, the classical order-three atoms and the δ^4-weighted panels, provided the head is a bounded overlap functional that is nonzero on ψ_0 and physical linear dependencies are quotiented out (PROVED). OPEN: identifying this physical weak pencil with the formal row-cleared recurrence of Table I, an explicit rate in N, and a classification of all formal solutions.

Numerically, for the plain 57-term recurrence at $\varepsilon = 1.704$, $|g_N(E_0)|$ falls from 1.8×10^{-4} ($N = 6$) to 6.1×10^{-8} ($N = 22$), roughly as N^{-5} over the tested range.

Raw-tail experiments and dichotomy. Classical scalar minimality compares a solution with independent ones through a ratio that tends to zero; it does not require geometric separation. For this recurrence, unit random data placed in shells $N+1$ to $N+3$ change the low shells by amounts of order 0.03 to 3 for $N = 10, \ldots, 22$, i.e. undamped (NUMERICAL). Geometric decay of the full plain Laguerre array would make F entire, contradicting the nonzero $R^2\ln R$ term (PROVED). Even Leaver's classical radial (Jaffé-type) recurrences have a double characteristic root 1 and only stretched-exponential separation [7, 8], so even there selection does not rest on a geometric dichotomy. We do not prove a basis-independent no-dichotomy theorem. The selecting notion here is a small tail in the form norm, not a coefficient ratio.

V. NUMERICS

Table II gives Ritz energies at fixed $\varepsilon = 1.704$, $E_0 = -2.903724377034119598\ldots$ (NUMERICAL, multiprecision arithmetic, all $E \geq E_0$). The log seed builds in the leading Fock logarithm, and min5 the order-two non-polynomial angular content; adding ψ_{30} gives only modest gains at comparable size. Log–log slopes against basis size steepen to about -8 to -15, while semilog curves remain convex. These finite data are consistent with fast algebraic convergence; no asymptotic rate is established. For comparison, Pekeris' recurrence at 525 functions gives $E - E_0 = 1.4\times10^{-8}$.

Remark. Fixing the Fock jets through $R^2\ln R$ exactly and reading the harmonic coefficient a_{21} (Liverts total-projection convention, $\psi_{00} = 1$, $\ln R$) through a global finite-part functional gives $a_{21} \approx 0.47674788$ at $N = 11$. Two bases agree to 3×10^{-9}, but the error scale is only empirical, about 10^{-8}; these are not certified digits.

VI. OPEN PROBLEMS

(1) Identify the physical weak pencil with the formal row-cleared recurrence (gauge, admissible tails), and classify all formal solutions. (2) Give an explicit rate in N for Theorem 5. (3) Find the least panel weight power in $\{1, 2, 3, 4\}$, and decide whether ψ_{30} lies exactly in a finite weighted span. (4) Prove minimality of the order-three ranks (12 classical, 13 per panel are candidates). (5) Extend to order four: does ψ_{40}, ψ_{41} close on a finite module, and is $\ln^p R$ nonzero for every p (CONJ)?

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